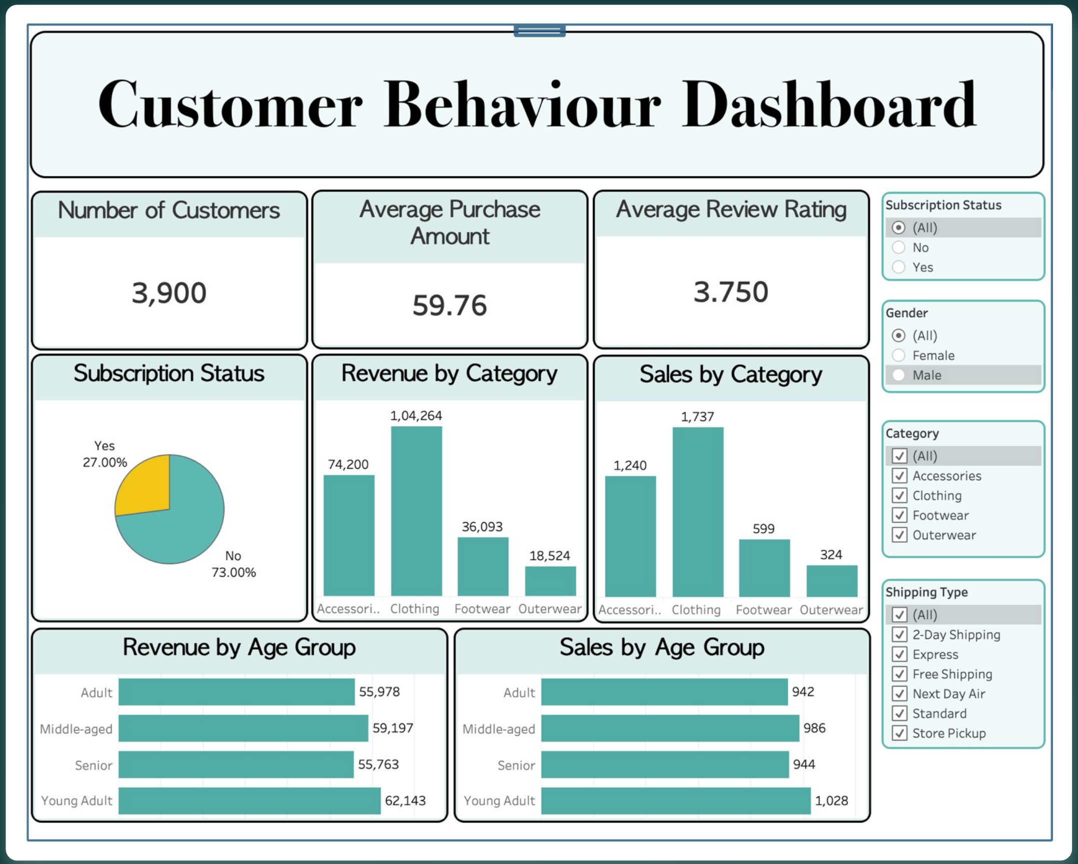
Customer Shopping Behavior Analysis

An end-to-end data analytics project

Python → PostgreSQL → Tableau



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Business Problem

THE KEY QUESTION

How can shopping data reveal trends to optimize marketing and product strategy?

Objective

Understand customer shopping behavior to lift sales, engagement, and long-term loyalty.

Focus Areas

- Discounts
- Reviews
- Seasonality
- Payment preferences
- Demographics

Approach

```
#create a column_age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q = 4, labels = labels)
```

```
df[['age', 'age_group']].head(10)
```

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

1

Python (pandas)

Cleaned data, imputed missing review ratings by category median, and engineered age_group and purchase-frequency features.

2

PostgreSQL

Loaded the data and answered 10 business questions using aggregates, subqueries, CASE, CTEs and window functions.

3

Tableau

Built a live-connected interactive dashboard with KPIs, category & age breakdowns, and cross-filters.

KPIs at a Glance

3,900
Total Customers

$59.76
Avg Purchase Amount

3.75 / 5
Avg Review Rating

27%
Subscribed (1,053)

As shown on the live Tableau dashboard

73% not subscribed

2,847 of 3,900 customers — the biggest growth opportunity.

Where the Revenue Comes From

Revenue by Category

- Clothing — $104,264 (1,737 orders)
- Accessories — $74,200 (1,240 orders)
- Footwear — $36,093 (599 orders)
- Outerwear — $18,524 (324 orders)

Revenue by Age Group

Young Adult $62,143 · Middle-aged $59,197 · Adult $55,978 · Senior $55,763

Insight Clothing is the core category (~45% of revenue and orders); revenue is spread evenly across all age groups.

The Demographics Story

```sql
--What is the total revenue generated by male vs. female customers?
select gender, sum(purchase_amount) as revenue
from customers
group by gender
```

Data Output Messages Notifications

	gender text	revenue numeric
1	Female	75191
2	Male	157890

SQL result — revenue by gender

Revenue by Gender

Male $157,890 (68%) vs Female $75,191 (32%)

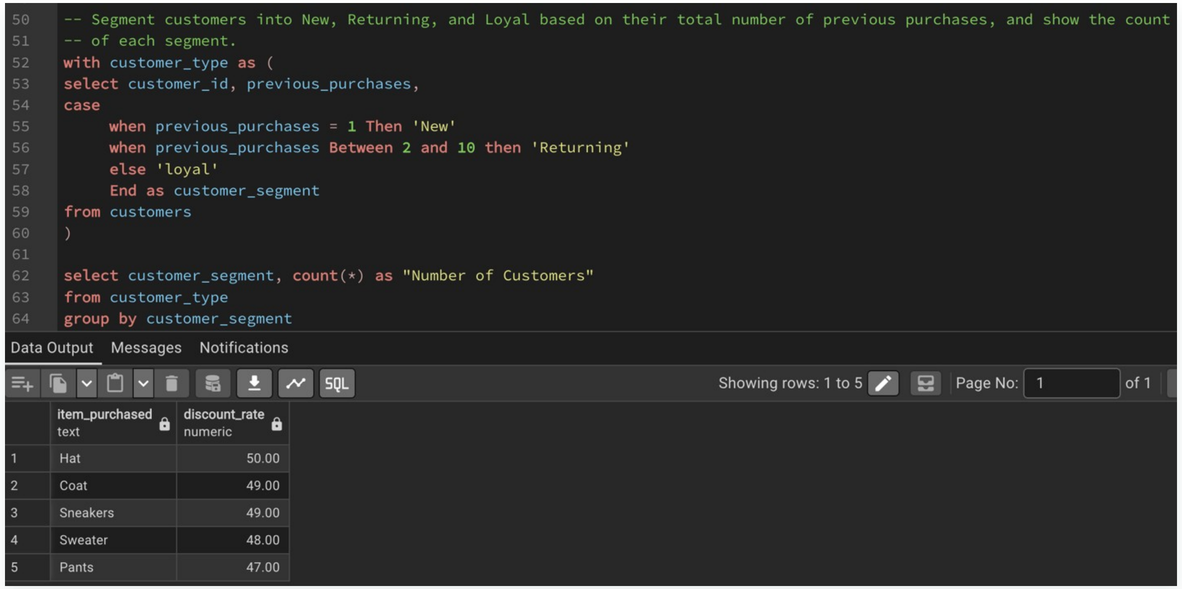
Per-Customer Spend

Male ≈ $59.5 · Female ≈ $60.2

Insight

The male lead is driven by a larger customer count — women actually spend marginally more per head. Acquiring more female customers is an untapped lever.

Subscription & Loyalty — Biggest Opportunity



```sql
-- Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count
-- of each segment.
with customer_type as (
select customer_id, previous_purchases,
case
     when previous_purchases = 1 Then 'New'
     when previous_purchases Between 2 and 10 then 'Returning'
     else 'loyal'
     End as customer_segment
from customers
)

select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Coat	49.00
3	Sneakers	49.00
4	Sweater	48.00
5	Pants	47.00

SQL result — customer segmentation

Key Figures

- Subscriber vs non-subscriber spend: $59.49 vs $59.87
- Segments: Loyal 3,116 · Returning 701 · New 83
- Repeat buyers (5+): 958 subscribe vs 2,518 don't (~28%)

Insight

80% of customers are loyal repeat buyers, yet only ~27% subscribe — and subscription doesn't lift spend. Converting loyal non-subscribers is the win.

Products, Discounts & Shipping

Top-Rated Products

- Gloves 3.86
- Sandals 3.84
- Boots 3.82
- Hat 3.80
- Skirt 3.78

Best-Sellers by Category

- Jewelry (Accessories)
- Pants & Blouse (Clothing)
- Sandals (Footwear)
- Jacket (Outerwear)

Most Discount-Reliant

- Hat 50%
- Coat 49%
- Sneakers 49%
- Sweater 48%
- Pants 47%

Shipping

- Express $60.48 vs Standard $58.46
- 839 customers spent above average while using a discount

Insight Some products only sell with heavy discounts (margin review advised); faster shipping correlates with modestly higher spend.

Recommendations

1 **Grow subscriptions**

73% aren't subscribed and it doesn't lift spend — redesign the offer with real value.

2 **Convert loyal non-subscribers**

Target repeat buyers (80% of base) who don't subscribe.

3 **Acquire more female customers**

Women are 32% of base but spend slightly more per order.

4 **Audit discount-reliant items**

Review margins on Hat, Coat, Sneakers (~49–50% discounted).

5 **Reinforce Clothing**

The core category (~45%); investigate Outerwear's low share.

6 **Feature top-rated products**

Promote best-rated & best-sellers; consider premium pricing.

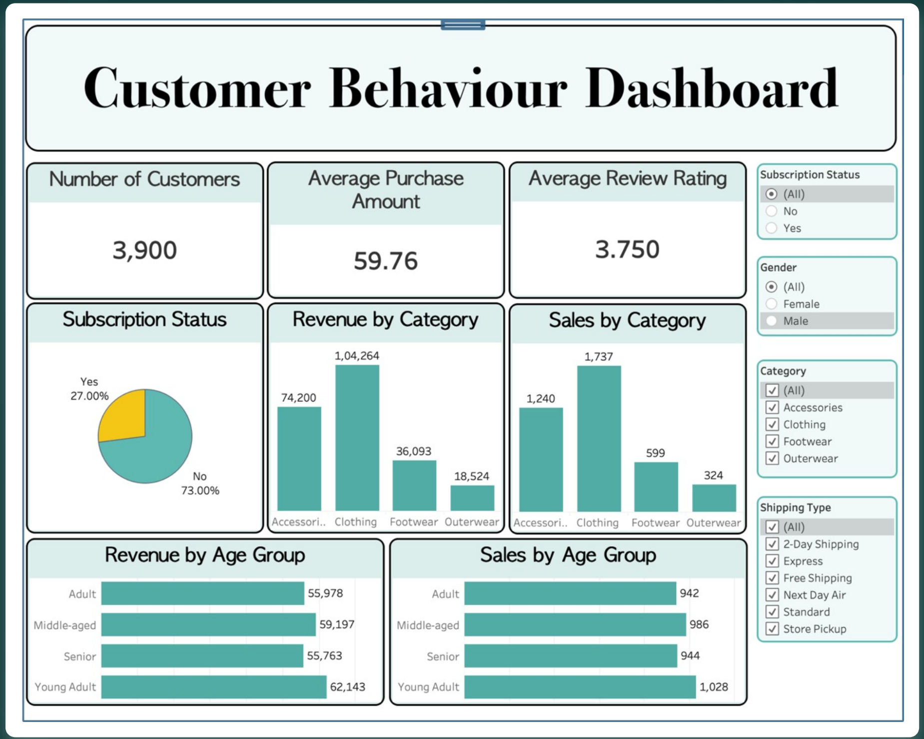
7 **Invest in faster delivery**

Express buyers spend modestly more.

Tools & Links

Tech Stack

- Python — pandas, Jupyter Notebook
- PostgreSQL — pgAdmin, SQLAlchemy + psycopg2
- Tableau — live database connection



Links

GitHub repository — *link added on publication*

Live dashboard / full report — *link added on publication*